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2025

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer **all** group as directed.*

Group—A

(Objective Type Questions)

1. Choose the correct answer : $2 \times 10 = 20$

(i) The main purpose of an Operating System is to:

A) Compile programs

B) Manage hardware and software resources

C) Design applications

D) Perform calculations

(ii) Which of the following is not a function of OS?

A) Process management

B) Memory management

C) File management

D) Word processing

(iii) The FCFS algorithm is particularly troublesome for:

A) Time Sharing systems

B) Multiprocessor systems

C) Operating systems

D) Multiproprogramming systems

(iv) Which of the following is an example of an Operating System?

A) MS Word

B) Linux

C) Python

D) Chrome

(v) If a process fails, most operating system write the error information to a?

A) New file

B) Another running process

C) Log File

D) None of the mention

(vi) Which scheduling algorithm is non-preemptive?

A) Round Robin

B) FCFS

C) Priority (Preemptive)

D) SJF (Preemptive)

(vii) Context switching occurs when:

A) Process terminates

B) CPU switches from one process to another

C) Interrupt occurs

D) System boots

(viii) Virtual memory is:

A) Secondary memory

B) RAM

C) Technique that uses disk as extension of RAM

D) Cache memory

(ix) To access the services of the operating system, the interface is provided by the

A) Library

B) System calls

C) Assembly instructions

D) API

(x) In operating system, each process has its own :

A) Open files

B) Pending alarms, signals

C) Address space, global variable

D) All of mentioned

Group – B

(Short Answer Type Questions)

Answer any **four** of the following : 5×4=20

2. State the main difference between logical and physical address space.

3. Explain the functions of an Operating System in detail.

4. Differentiate between batch-system and time sharing system.

5. Explain the structure of an OS.

6. How does swapping result in better memory management?

Group – C

(Long Answer Type Questions)

Answer any **two** of the following : 15×2=30

7. Describe in details the different types of Operating Systems: Multiproprogramming, batch and time sharing.

8. Explain Preemptive Priority Scheduling Algorithms in detail.

9. Give the view of Operating system as a Resource Manager.

———— X ————